

Problem Set 7

Econ 5253 — Spring 2026

Sumedha Ray

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Question 6: Summary Statistics and Missing Data

Table 1: Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
logwage	1,669	1.63	0.39	0.005	2.26
hgc	2,229	13.10	2.52	0	18
tenure	2,229	5.97	5.51	0.00	25.92
age	2,229	39.15	3.06	34	46

The `logwage` variable is missing for 25.12% of observations. I believe `logwage` is most likely **Missing Not at Random (MNAR)**. Women with very low or very high wages may be systematically less likely to report earnings, meaning the probability of missingness depends on the unobserved wage value itself. This violates both MCAR and MAR.

Question 7: Imputation Methods and Returns to Schooling

I estimate the following model under four imputation strategies:

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{hgc}_i + \beta_2 \text{college}_i + \beta_3 \text{tenure}_i + \beta_4 \text{tenure}_i^2 + \beta_5 \text{age}_i + \beta_6 \text{married}_i + \varepsilon_i \quad (1)$$

The true value of $\hat{\beta}_1 = 0.093$.

All four methods underestimate the true $\hat{\beta}_1 = 0.093$, but the degree of attenuation differs across methods. Mean imputation performs worst ($\hat{\beta}_1 = 0.0497$), which is expected: replacing missing wages with the sample mean removes all covariance between log wages and the regressors, attenuating every slope coefficient toward zero. Complete cases does better ($\hat{\beta}_1 = 0.0624$), though it relies on the MCAR assumption which is likely violated here. Predicted value imputation ($\hat{\beta}_1 = 0.0571$) is derived directly from the complete cases model,

Table 2: Returns to Schooling Under Different Imputation Methods

	<i>Dependent variable:</i>		
	Complete Cases	Log Wage Mean Imputation	Predicted Values
	(1)	(2)	(3)
Years of Schooling (hgc)	0.0624*** (0.0054)	0.0497*** (0.0043)	0.0571*** (0.0044)
College	0.1452*** (0.0345)	0.1682*** (0.0257)	0.1453*** (0.0262)
Tenure	0.0495*** (0.0052)	0.0382*** (0.0039)	0.0407*** (0.0040)
Tenure ²	−0.0016*** (0.0003)	−0.0013*** (0.0002)	−0.0013*** (0.0002)
Age	0.0004 (0.0027)	0.0002 (0.0022)	−0.0007 (0.0022)
Married	−0.0220 (0.0177)	−0.0268** (0.0136)	−0.0248* (0.0139)
Constant	0.5336*** (0.1460)	0.7076*** (0.1158)	0.6869*** (0.1179)
Observations	1,669	2,229	2,229
R ²	0.2084	0.1473	0.1922
Adjusted R ²	0.2055	0.1450	0.1900
Residual Std. Error	0.3436 (df = 1662)	0.3085 (df = 2222)	0.3141 (df = 2222)
F Statistic	72.917*** (df = 6; 1662)	63.973*** (df = 6; 2222)	88.130*** (df = 6; 2222)

*Note:*True value of $\hat{\beta}_1 = 0.093$. MICE estimate: $\hat{\beta}_1 = 0.0583$.

so it mechanically preserves the covariate relationships but inflates precision by treating imputed values as observed.

The MICE estimate ($\hat{\beta}_1 = 0.0583$) is the most principled of the four. By drawing multiple plausible values for each missing observation via predictive mean matching and pooling across five imputed datasets, MICE correctly propagates uncertainty from the missing data into the standard errors. That said, all estimates remain below the true value of 0.093, suggesting the data are likely MNAR and that no imputation method fully recovers the true parameter without modeling the missingness mechanism directly.

Question 8: Project Progress

My project, which I am developing as a potential job market paper, asks whether economists who study sensitive topics—such as race, religion, and discrimination—face systematically worse labor market and career outcomes than observationally comparable candidates working on conventional research topics. The core identification leverages variation in dissertation topic (classified using JEL codes and keyword matching from ProQuest Dissertations & Theses) to ask three related questions: (1) does working on sensitive topics reduce the probability of obtaining an academic position, conditional on candidate quality and program rank? (2) do researchers on these topics shift their agendas around the tenure threshold, consistent with pre-tenure conformity pressure or post-tenure intellectual liberation? and (3) to what extent do observed patterns reflect demand-side discrimination versus rational supply-side self-censorship? My advisor has suggested framing the topic selection as a discrete choice/Roy model to account for selection into sensitive research areas based on demographics and program characteristics. I am currently building the data pipeline using ProQuest, RePec, and department placement pages.